01/29

Wednesday, 29 January 2025 9:31 AM

Midterms → only conceptual qs

Final → algorithm

Roots:

What method can guarantee root?

↳ Bisection (but have to cross axis i.e. change in value from + to −)

↳ Newton-Rhapson (N-R) (initial guess must be w/in convergence limit)

↳ If slope is zero (it will eventually approach it but never reach)

↳ Modified N.R. to deal w/ this

Ax=b

- Why pivoting?
  - ↳ Avoid division by zero & avoid floating / round-up error [trying to find the biggest number]
  - ↳ equilibration → division by biggest term in equation ↓ to compare each equation
- Checking solution: residual norm & iterative

- Diverging solutions for J & G-S methods?
  - ↳ Expand radius of convergence

PDE's

- ↳ Linear, almost linear, quasi-linear, non-linear
  - Highest order must be non-linear
  - Linear: when a no-directive term is non-linear
  - Quasi-linear: higher-order is linear but lower order is non-linear

- ↳ Second order PDE classification
  - + Second order coefficients matrix
  - + 5 different forms: parabolic, hyperbolic, elliptical
    - parabolic: narrow, broad

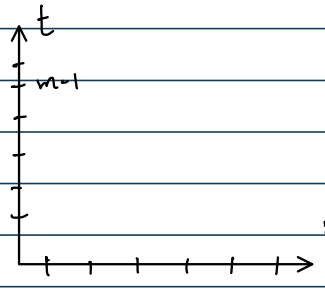
$U_t \Rightarrow \frac{\partial u}{\partial t}$

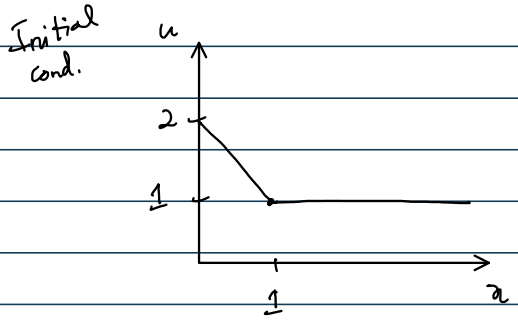
$UU_x \Rightarrow \frac{\partial u^2}{\partial x^2}$

Conservative vs. non-conservative form

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$$

IC: $u \begin{cases} 2-x & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$





Multiple solns
Create "shock waves"
↓
discontinuity
(weak solution)

